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Quality seedling systems for sustainable urban agriculture: evidence from four rapidly urbanizing cities in Africa and Asia

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Rapid urbanization is placing increasing pressure on cities to deliver affordable, nutritious, and resilient food systems, yet vegetable seedling systems remain a neglected component of urban agriculture. This study assessed urban and peri urban vegetable seedling systems in four rapidly growing cities Addis Ababa (Ethiopia), Nairobi (Kenya), Dhaka (Bangladesh), and Metro Manila (Philippines) to identify key constraints, opportunities, best practices, and policy lessons. Using a comparative multi city case study approach, data were collected from 154 stakeholders, including urban farmers, nursery operators, seedling suppliers, agro input dealers, and government officers, complemented by secondary literature. Across the four cities, urban vegetable production was dominated by small scale, intensive systems cultivating predominantly leafy vegetables, while seedling systems remained underdeveloped, characterized by informal nursery operations, weak quality assurance, open field propagation, limited technical capacity, and fragmented market linkages. These constraints contributed to widespread reliance on direct seeding and low-quality planting material despite strong demand for quality seedlings. Comparative analysis identified opportunities to strengthen commercial and community nursery enterprises through improved propagation technologies, certification systems, digital marketing, commercialization of indigenous vegetable seedlings, and stronger public private partnerships. The study suggests that integrating seedling enterprises into urban food system strategies, supported by improved financing, extension, and quality assurance, could substantially enhance the productivity, resilience, and sustainability of urban vegetable production. As one of the first cross continental assessments of urban vegetable seedling systems in low and middle income countries, the study fills an important evidence gap and positions seedling systems as a critical, yet overlooked, component of resilient and nutrition sensitive urban food systems.

KEYWORDS

climate smart, food security, garden, resilience, urban food systems, vegetables

1 Introduction

Urbanization is one of the defining trends globally. Over half the world's population currently lives in cities, which is expected to increase to 68% by 2050 (UN Habitat, 2022). Cities in Africa and Asia are expanding particularly rapidly, placing increasing pressure on urban food systems to provide sufficient, affordable, and nutritious food. Although food insecurity has traditionally been viewed as a rural challenge, an estimated 77% of people experiencing moderate or severe food insecurity now live in urban and peri-urban (UPU) areas (Battersby et al., 2024). Urban populations face a growing double burden of malnutrition, characterized by the coexistence of micronutrient deficiencies and rising rates of overweight, obesity, and diet-related non-communicable diseases (FAO et al., 2023a). Fruits and vegetables could address both challenges because they provide essential vitamins, minerals, and dietary fibre while reducing the risk of chronic diseases. Yet, in many low-income countries, meeting the recommended daily intake of fruits and vegetables would consume nearly one-third of urban households income (Miller et al., 2016).

Urban and peri-urban agriculture has emerged as an important strategy for strengthening urban food security and nutrition, livelihoods, and resilience while shortening food supply chains (Orsini et al., 2013; Hamilton et al., 2014; Weidner et al., 2019). Producing more food in and near cities could increase the availability and affordability of fresh and nutritious produce while generating broader social and environmental benefits, including greener urban landscapes, community cohesion, and improved household wellbeing (Orsini et al., 2013; Garcia et al., 2018; Schreinemachers et al., 2025). However, the potential of UPU agriculture to deliver these benefits is underexploited. The productivity of UPU agriculture is often low, the safety of urban produce is often questionable, while scarce resources, such as space and water, are not optimally utilized (Maxwell et al., 1998; Orsini et al., 2013).

One critical but often overlooked component of UPU food systems is the availability, accessibility, and quality of vegetable seedlings (Coyne et al., 2024). Although quality seedlings improve crop establishment, seed-use efficiency, productivity, and resilience, access to reliable planting material remains constrained in many low- and middle-income countries because commercial nursery systems are poorly developed, quality assurance is weak, and informal input markets predominate (Orsini et al., 2013; Schreinemachers et al., 2018; Weidner et al., 2019; Gallegos-Cedillo et al., 2024). At the same time, UPU agriculture presents considerable opportunities for specialized seedling enterprises, as vegetables are commonly grown on small plots, rooftop gardens, and containers where quality seedlings can maximize space use, shorten production cycles, improve productivity, and reduce pest and disease losses. Despite their importance, research and policy have focused largely on seed systems, with comparatively little attention to seedling production, nursery enterprises, quality assurance, market development, and institutional support, particularly in rapidly urbanizing cities where informal nursery systems predominate (Orsini et al., 2013; Weidner et al., 2019; Coyne et al., 2024).

This study aimed to assess the status of seedling systems in cities of lower-income countries and to identify key bottlenecks and opportunities towards improving the productivity and safety of UPU agriculture. The goal is to provide a detailed understanding of how seedling systems currently function, the challenges they face, and how they can be transformed to support resilient and productive urban food systems. This was conducted using a case study approach in four

large and rapidly growing cities in Africa and Asia: Addis Ababa, Nairobi, Dhaka, and Metro Manila.

In Addis Ababa, food security initiatives and household gardens are promoted in response to rapid urbanization and changing climate patterns. Nairobi, a dynamic hub in East Africa, balances informal and formal UPU farming, increasingly integrating vertical farming and protected cultivation. In Dhaka, one of the world's largest cities, rooftop gardening is popular, creating a strong demand for vegetable seedlings. In Metro Manila, urban farming has been endorsed by the national and local governments as a resilience measure against food supply shocks and rising food prices. Across all four cities, vegetables are central to UPU farming due to their short growth cycles, nutritional value, perishability and suitability for growing in small spaces. However, seedling systems are critical for sustaining UPU vegetable farming and face major constraints related to quality, access, institutional support, and market integration.

The analysis draws on scoping studies conducted in each city, supplemented with comparative insights and real-world examples of innovations. It presents both city-specific findings and cross-cutting themes, leading to practical recommendations for reform, investment, and capacity building.

This study makes an important contribution to the literature on sustainable urban food systems. It provides one of the first multi-city, cross-continental assessments of vegetable seedling systems in low- and middle-income countries, filling an important gap in a literature that has largely concentrated on seed systems in rural and peri-urban contexts. By drawing on primary stakeholder data from four rapidly urbanizing cities, the study generates actionable insights for practitioners, policymakers, and investors seeking to strengthen urban agriculture. The framing of seedling systems as a critical but neglected link in the urban food system also offers a useful conceptual contribution that complements existing discussions of urban food system resilience and inclusive agribusiness (Battersby et al., 2024; Claereboudt et al., 2024; Coyne et al., 2024).

2 Materials and methods

2.1 Site selection

The CGIAR Resilient Cities Initiative, now Food Frontiers and Security (2022–2024)¹ targeted six cities across Africa, Asia, and Latin America, selected for their diversity in geography, governance models, and UPU agricultural activity levels. Four of these cities were selected for the current study: Nairobi, Kenya (Obebo and Coyne, 2023); Addis Ababa, Ethiopia (Adimassu et al., 2024); Dhaka, Bangladesh (Alam et al., 2023); and Metro Manila, Philippines (Arro, 2023). These cities vary in population size, from ~6 million for Nairobi and Addis Ababa to 15 million for Metro Manila and 25 million for Dhaka, the world's fourth-largest city (Table 1). Dhaka is expected to grow to 31 million by 2035, while Nairobi and Addis Ababa are also experiencing rapid expansion. All four cities have a large proportion of their population in informal residential settlements, ranging from 36% in Metro Manila to 64% in Addis Ababa (UN Habitat, 2022). The prevalence of food insecurity is also high: 44% in Metro Manila, 56% in Addis Ababa, and 70% in Nairobi.

The study was designed as an exploratory synthesis rather than a comparative analysis. The goal was not to rank cities or draw direct

TABLE 1 Descriptive statistics of the four cities included in the study.

Indicator	Nairobi, Kenya	Addis Ababa, Ethiopia	Metro Manila, Philippines	Dhaka, Bangladesh
Population (million) ¹	5.8	6.0	15.2	24.7
Annual population growth (%) ²	3.6	4.4	1.5	3.1
Child stunting (% of children 0–59 months) ³	16.3	12.0	NA	17.0
Population in informal settlements (%) ⁴	40.5	64.3	35.9	51.5
Built-up area per capita (m ² /person) ²	34.7	52.1	52.3	17.4
Green area per capita (m ² /person) ⁵	4.1	NA	41.3	1.4
Green area in city (%) ⁵	4.2	NA	44.7	5.1
Prevalence of food insecurity (%) ⁶	69.5	56.2	43.8	NA
Mean air temperature (C) ⁷	18.9	17.0	26.6	26.0
Annual precipitation (mm) ⁷	920	1175	2614	1944

¹Population in Urban Agglomerations 2025. <https://guo-un-habitat.maps.arcgis.com/sharing/rest/content/items/9ca4df2304ee457982eb04d7d19dbf11/data>

²Data refer to 2010–2020. UN Habitat Urban Indicators Database (January 2025). <https://data.unhabitat.org/pages/spatial-growth-of-cities-and-urban-areas>

³<https://guo-un-habitat.maps.arcgis.com/sharing/rest/content/items/b536def3118649d4967242ab4cddb6d6/data>

⁴Data refer to 2022 and to the entire urban population in the country. <https://guo-un-habitat.maps.arcgis.com/sharing/rest/content/items/abcc290e8aa940f2a4deade7a338c6d6/data>

⁵2020 <https://guo-un-habitat.maps.arcgis.com/sharing/rest/content/items/cc37148f95304783a6ccfad8a9124a9d/data>

⁶<https://guo-un-habitat.maps.arcgis.com/sharing/rest/content/items/cbe724c101774ee2aa74cdd940fffc7a/data>

⁷<https://climateknowledgeportal.worldbank.org/>

comparisons across contexts, but to document the diversity of seedling system configurations and identify systemic patterns and opportunities that could inform policy and investment decisions across comparable rapidly urbanizing settings. Cities in Africa and Asia were selected because these are the regions experiencing the fastest rates of urban population growth, and where the gap between seedling system capacity and urban food demand is most acute.

2.2 Approach and data collection

We developed a common study protocol to ensure comparable data collection across four cities: Nairobi (Kenya), Addis Ababa (Ethiopia), Dhaka (Bangladesh), and Metro Manila (Philippines). The protocol focused on four thematic areas: (i) the status of UPU vegetable production, (ii) market demand for vegetable seedlings, (iii) agronomic aspects of quality seedling production, and (iv) policy and institutional support. Each thematic area was guided by five to eight research questions. The city assessments were conducted as standalone scoping studies (Obebo and Coyne, 2023; Alam et al., 2023; Arro, 2023; Adimassu et al., 2024), which form the basis of the synthesis presented here.

Data collection combined a desk review with primary field surveys. The desk review synthesized published and grey literature identified through Google Scholar, CAB Abstracts, and CGIAR institutional repositories using keywords related to urban agriculture, vegetable seedling systems, nurseries, and urban food systems. Additional sources included city-specific policy documents, technical reports, and project publications. Literature was selected based on its relevance to urban and peri urban vegetable production, seedling systems, nursery enterprises, policy frameworks, and institutional support in the

four study cities and comparable low and middle income country contexts. Primary data were collected through structured interviews and surveys with stakeholders across the vegetable seedling value chain, including nursery operators, vegetable growers, government officials, agro-input dealers, seed retailers, and representatives of development organizations. Between 2022 and 2024, a total of 154 stakeholders were interviewed across the four case study cities, with the composition of respondents summarized in Table 2.

Documentary evidence and interview data were analyzed using a thematic synthesis approach. Data were analyzed within each city to identify patterns related to seedling system characteristics, constraints, opportunities, and institutional arrangements. This process generated four cross-cutting themes: (i) the status of urban agriculture, (ii) characteristics of urban vegetable seedling systems, (iii) opportunities and constraints, and (iv) policy and institutional support. Given the exploratory nature of the study, the analysis was descriptive and did not involve inferential statistical methods.

3 Results

3.1 Status of urban and peri-urban agriculture

The four cities have large populations, requiring enormous quantities of vegetables daily. Even the smaller cities, Nairobi and Addis Ababa, require well over 1 million kilograms of vegetables a day to feed their populations. There are no reliable data on urban production, however, with just a fraction of the daily vegetable supply likely

TABLE 2 Summary of primary data collection across the four case study cities ($n = 154$).

City	Study participants (n)	Study area
Nairobi, Kenya	55: 50 commercial seedling nursery operators and 5 key informants from the County Government, Mazingira Institute, RUAF, Nairobi and Environs Food Security, Agriculture and Livestock Forum, and African Population and Health Research Center.	Nairobi and peri-urban around Kiambu and Machakos regions
Addis Ababa, Ethiopia	36: Vegetable seedling producers and growers, complemented by key informant interviews with government and non-governmental organizations.	Bole, Lemi Kura and Akaki Kaliti sub-cities
Dhaka, Bangladesh	31: Urban gardeners, seedling retailers, seed producers, nursery operators, government officials, and representatives of research and development organizations.	Dhaka North and Dhaka South City Corporations
Metro Manila, Philippines	32: Urban gardeners, commercial farmers, nursery operators, and government officers.	Metro Manila and the peri-urban provinces of Bulacan, Pampanga and Rizal

produced in or near the city, the bulk comes from rural areas. Metro Manila and Dhaka are located relatively distant from the main vegetable-producing areas (while Nairobi and Addis Ababa have shorter supply chains). Each of the four cities has a vibrant urban agriculture, as described in the following and summarized in Table 3.

In Addis Ababa, vegetable production is vibrant but largely characterized by informal practices. Farmers cultivate a wide range of vegetables, with a strong preference for leafy greens. Nearly all respondents ($n = 36$) produced spinach (97.2%) and kale (97.2%), while 94.4% grew lettuce. Other commonly cultivated crops included head cabbage (72.2%), onions (63.9%), and tomatoes (52.8%). Urban agriculture is actively supported by the Addis Ababa Urban Agriculture Commission through home gardens and community farming initiatives, although production remains predominantly small-scale and geared toward household consumption, with relatively few commercial enterprises. Survey respondents ($n = 32$) were predominantly male (77.8%), with an average age of 36.4 years and an average of 8.4 years of experience in vegetable seedling production. Key constraints identified by producers included pest and disease pressure, the high cost and limited availability of quality inputs, and inadequate extension and technical support.

In Nairobi, urban agriculture comprises small-scale household production alongside larger commercial operations. The Jomo

Kenyatta University of Agriculture and Technology, and Food and Agriculture Organization (2018) food systems assessment for Nairobi found that vegetable farming accounts for 46% of agricultural enterprises in the city. Smallholders typically cultivate plots averaging 0.125 acres, while commercial producers increasingly use protected cultivation, including greenhouses. Kale, spinach, tomatoes, and onions dominate production, with African Indigenous Vegetables remaining relatively underutilized. Consistent with these findings, our survey of commercial nursery operators ($n = 50$) showed that 88% propagated kale, 84% spinach, 80% onions, 76% cabbage, and 72% lettuce, reflecting strong market demand for these crops. Nursery ownership was predominantly male (42% male-owned and 24% jointly owned), with most operators aged over 40 years. Urban agriculture is supported by Nairobi City County and development partners through extension, training, and food systems initiatives. However, limited access to land, weak extension services, high input costs, and pest and disease pressure remain major constraints.

In Manila, Philippines, UPU vegetable production is dominated by community gardeners and individual home gardens. Community gardeners ($n = 15$) primarily produce short-duration leafy vegetables, including lettuce, bok choy, Chinese cabbage, spinach, and spring onions, while commercial farmers cultivate similar crops on leased plots ranging from 200 to 2,400 m². Demand for seedlings is greatest for tomato, eggplant, pepper, bitter melon, and lettuce, although many commercial farmers prefer direct seeding for leafy vegetables. Urban agriculture is supported through the National Urban Agriculture Program and local government initiatives that promote community gardens and provide free vegetable seedlings and extension support, particularly to low- and middle-income households. Commercial nursery operators, located mainly in peri-urban Bulacan, supply seedlings largely to rural markets, with relatively limited demand from urban producers. Key challenges identified by respondents included pests and diseases, particularly damping-off, poor-quality growing media, unreliable water supply, flooding associated with typhoons, and the limited availability of quality commercial seedlings.

In Dhaka, urban and peri-urban vegetable production is dominated by household gardens, particularly rooftop gardens, with relatively few commercial farms except in peri-urban areas such as Gazipur. Survey respondents ($n = 31$) indicated that rooftop gardens produce a diverse range of vegetables, including gourds, cucumber, okra, eggplant, tomato, chili, onion, cauliflower, radish, Indian spinach, and amaranth, alongside fruit trees such as papaya, lemon, and guava. Women manage most rooftop gardens, which are typically owned by middle- and upper-income households and primarily aim to meet household food needs. Urban agriculture is supported by government agencies and non-governmental organizations through rooftop gardening initiatives, training, and awareness programs, while social media platforms increasingly facilitate seedling marketing and knowledge exchange. Key constraints included pest and disease pressure, limited technical knowledge, high input costs, and restricted access to rooftop space and water, particularly for households living in rented accommodation.

3.2 Characterizing seedling systems for urban and peri urban agriculture

Vegetable producers in each of the four cities depend on seed rather than seedlings because the supply of seedlings is limited. While seedling suppliers are present in each city, the widespread issue of

TABLE 3 Characterization of urban agriculture in the four target cities.

Characteristic	Nairobi, Kenya	Addis Ababa, Ethiopia	Metro Manila, Philippines	Dhaka, Bangladesh
Where do most vegetables come from?	From peri-urban areas, but also within Nairobi. The supply falls short of demand.	From Addis Ababa and the surrounding peri-urban areas.	From afar; production around Metro Manila is minimal. Vegetables are relatively costly.	From afar; production around Dhaka is limited.
What are the main types of urban food producers?	Homestead production, small-scale production on public/private lands, and commercial greenhouses. Micro gardening is emerging.	Commercial producers to community gardens, smallholder family farms.	Community and individual home gardens.	Rooftop and land-based gardens.
What is the primary purpose?	Mostly commercial, but also household consumption.	Own household consumption, few for commercial purposes.	Own household consumption; commercial operations are rare.	Own household consumption; commercial operations are rare.
Who engages in urban agriculture?	Mainly men, often seniors. Few women and youth.	Mainly men with an average age of 36 years.	Women, many are seniors.	Mostly women, rooftop gardeners are well-off.
What are the key constraints?	Lack of land and extension services, pests and diseases, costly inputs.	Pests and diseases, limited and costly inputs, lack of extension services, poor knowledge.	Pests and diseases	Pests and diseases, poor knowledge, costly inputs.

poor-quality vegetable seedlings affects all locations (Table 4). This poor quality mainly arises from the lack of greenhouses and quality sowing media, as propagators often sow in open fields to minimize costs.

Urban food producers in Addis Ababa mainly use seed rather than seedlings. Producers source seeds from both formal and informal channels. Vegetable seeds can be purchased from shops and open markets throughout the city. Saving their own seed is crucial for growers aiming to avoid the high costs of certified seed. While commercial vegetable producers typically grow their own seedlings, their nurseries are often located in open fields without containers. This approach is not ideal, as soil exposure can lead to soil-borne diseases and root damage during transplantation. These so-called 'bare root' seedlings are also prone to transplant shock, unlike propagated tray seedlings, which have a strong and vigorous root mass contained within a potting media and are 'ready to go'. The supply of vegetable seedlings from commercial sources is limited, though options are available. These include seedlings of various leafy vegetables and herbs, as well as saplings of fruit trees like apples, guava, and avocado. Commercial vendors sell seedlings in pots or plastic bags. Grafting is not used for vegetable seedlings, only for select fruit saplings.

The city of Nairobi has several private commercial nursery operators, although most urban food producers use seed rather than seedlings. Seed supplies are reliable and the seed is of decent quality, but there are also seeds from unknown sources. Most nurseries are small-scale. There are approximately 300 nurseries that are formally certified, but there are many more informal operators. There are also a few larger-scale operations. Small-scale nurseries typically sell saplings of fruit trees, ornamentals, and herbs, alongside vegetable seedlings. Popular vegetable seedlings include kale, spinach, tomato, onion, and cabbage, sold mainly through informal markets, roadside sellers, and occasional supermarket outlets. Customers also visit the nurseries to buy seedlings. The use of plastic trays is common. Grafting is not

practiced on vegetables. Large-scale operators produce seedlings of good quality, as they are produced from quality seeds in greenhouses using non-soil media. However, over 90% of seedlings come from small nurseries that are produced in the open field. Market linkages between nurseries and vegetable producers are underdeveloped, resulting in mismatches between supply and demand.

Seed is also the main type of vegetable propagation material in Metro Manila. However, the government recognizes the importance of seedlings and, through the National Urban and Peri-urban Agriculture Program (NUPAP, 2025) as well as local government units, provides urban gardeners with vegetable seedlings free of charge. Popular seedlings include eggplant, tomato, pepper, bitter melon, and leafy vegetables. The seedlings are not grafted and the overall quality of the government-supplied seedlings is generally poor, often resulting in weak, disease-prone plants. Plastic seedling trays are commonly used. Most commercial nurseries operate in rural areas outside Metro Manila and primarily serve rural-based commercial producers rather than urban gardeners. These commercial nurseries tend to be specialized in a few crops, such as tomatoes, eggplant, hot pepper, papaya and grafted bitter melon.

Gardeners in Dhaka also rely on seed rather than seedlings for vegetable production. Vegetable seeds are widely available in small packets sold at retail shops and by mobile vendors, but their quality is often perceived as unreliable. As a result, customers tend to prefer seeds of hybrids, especially those from trusted companies. Vegetable seedling nurseries are mainly located in the peri-urban areas of Dhaka, such as Gazipur and Narayanganj. Agronomic practices among nurseries are inconsistent: while some operators sterilize soil and manage pH level, most of them lack formal training and face difficulties in controlling pests and diseases. Most nurseries grow seedlings in open field conditions rather than in enclosed or protected nurseries. Seedlings are typically sold in non-biodegradable polybags, and seedling trays are rarely used. There are no grafted vegetable

TABLE 4 Characterization of urban seedling systems in four target cities.

Characteristic	Nairobi, Kenya	Addis Ababa, Ethiopia	Metro Manila, Philippines	Dhaka, Bangladesh
Nursery operators	Private commercial nursery operators.	Commercial farmers produce their own seedlings in open field or pots. Some use greenhouses.	Government-run nurseries. Commercial nurseries focus on rural areas.	Mostly small-scale, low-technology, non-licensed (informal)
Types of seedlings sold	Mixed nurseries for fruit trees, vegetables, herbs, flowers and forest trees. Traditional leafy vegetables are common.	Wide range of vegetable seedlings of many leafy vegetables and herbs. Fruit tree saplings of apple, guava, avocado and plum.	Eggplant, tomato, pepper, bitter melon, leafy vegetables such as lettuce, papaya, squash, okra, and bottle gourd	Papaya, tomato, chili pepper, cucumber, beans, coriander leaves, radish, eggplant, carrot, pumpkin, and others
Packaging	Non-recyclable trays and polythene bags. Some renewable, low-cost options for fruit trees.	Pots and plastic bags, but often no packaging materials used.	Mostly non-recyclable seedling trays.	Mostly polybags, trays are uncommon.
Seedling quality	Medium to poor, but high for commercial nurseries.	Medium to poor	Medium to poor	Medium to poor
Grafted seedlings	Only for fruit tree saplings.	Slowly being adopted for fruit saplings.	Not available or rare	Not available
Availability of seedlings	Commercial nurseries supply seedlings <i>ad hoc</i> for small-medium sized customers. Large orders need to be ordered in advance.	Most seedlings are home grown.	Government organizations provide vegetable seedlings for free. Commercial supply is minimal.	Nurseries sell at wet markets, weekly markets, and sometimes online.
Availability of seed	Good availability from private and public stockists, but seed from unregulated sources and own seed is also used.	Found in urban shops and local open markets	Generally available from agricultural supply shops	Vegetable seeds in minipacks are available
Seed quality	Generally reliable	Unreliable	Generally reliable	Considered unreliable

seedlings. Nurseries supply seedlings to retailers in urban markets, including wet markets or weekly markets, and some have begun selling online. The seedlings include various crops like papaya, chili pepper, cucumber, beans, coriander, radish, eggplant, carrot, and pumpkin.

3.3 Opportunities and constraints for urban and peri urban agriculture

Although vegetables are cultivated by growers in urban and peri-urban areas for sale to market, a large component of urban food producers in the four cities are gardeners who grow food for their own household use. Gardeners are not professional farmers, spending relatively little time in their gardens each day. These gardeners aim for a variety of vegetables so they can harvest regularly using small plots that are often just a few square meters. Therefore, they need just a few plants of each type. Buying a seed package is not very appealing because a 10-gram minipack of tomato seed can contain up to 3,000 seeds, which is excessive for their needs. Gardeners often fail to establish a plant from a seed due to irregular watering or high temperatures. Vegetables are much more resilient after the seedling stage, once the

roots have developed and the plant has more leaves. Supplying urban gardeners with propagated tray seedlings, therefore, presents a more durable mechanism for productive home gardening, and also a business opportunity. We next review these opportunities, as well as challenges, for each of the four cities (Table 5).

Stakeholders in Addis Ababa reported a high demand for seedlings among farmers in the city. The city government also supports urban agriculture, although national priorities still focus on staple grain production. There is potential for public-private sector collaboration in seedling production, as some minimal public support could help reduce risks for investors. A key challenge is to keep seedling costs low and maintain quality, while ensuring a timely supply to urban producers. Using greenhouses, high-quality seeds, clean soil-less media, and clean water is essential for producing high-quality seedlings. Seedling production offers opportunities for employment creation for women and youth.

In Nairobi, stakeholders also believe there is a high potential demand for quality seedlings among urban food producers. Technologies such as greenhouses, clean soil-less media, and plastic trays are readily available in Nairobi, and a large population of women and youth is interested in working in nurseries, along with donors

TABLE 5 Key opportunities and constraints to developing urban seedling systems.

Nairobi, Kenya	Addis Ababa, Ethiopia	Metro Manila, Philippines	Dhaka, Bangladesh
Opportunities:			
High demand for quality seedlings. Leveraging on ICT to expand market information on seedlings to increase uptake. Existing network of potential seedling outlets. Government policies for seed certification standards. Strong support for youth inclusion. Rising demand for safer vegetables.	Perceived high potential demand for seedlings. Price monitoring system to provide farmers with seedling information. Local distribution channels for quality products. Public-private collaborative efforts to enhance vegetable production. Initiatives that support women and youth. Growing interest in healthier diets.	Unmet demand for quality seedlings. Increased UPU seedling sales through commercial nurseries. Potential for community farms to sell seedlings. Strong policy support for urban agriculture. Increased supply of soil-less potting media.	Unmet demand for quality seedlings. Marketing seedlings through online platforms. Gardeners prefer seedlings with a root ball, which are not always available. Introduction of biodegradable plastics.
Common opportunities: Financing mechanisms to support quality seedling propagation and use of other modern technologies such as advanced culturing techniques, infrastructure business development, etc. and commercialization and scale-up of indigenous leafy vegetables.			
Constraints:			
Ease of access to bare-root seedlings. Lack of adequate and quality water. Land availability Credit to invest in technologies. Unfavorable business environment for seedlings. Low access to market information.	Low demand for more costly high-quality seedlings. Contamination and shortage of water. Insecure land tenure and lack of land. Unregulated movement of livestock. Low level use or investment in greenhouse systems. National interest in promoting cereal production.	Low quality of government-supplied seedlings.	Women's access to markets to buy seedlings. Increasingly scarce land for nurseries.
Common constraints: Limited access to government extension and training services in urban areas. Pests and diseases, especially difficult to manage soil-borne issues.			

eager to support job creation. There is an opportunity for nursery operators to utilize online platforms and phone apps to connect with customers. Larger nursery operators could also adopt improved technologies, such as grafting or tissue culture to add value. However, taking advantage of these opportunities requires building capacity. Government extension services have generally limited access to vegetable growers. Access to financial services is another major obstacle that needs to be addressed.

Stakeholders in Metro Manila also believe there is an unmet demand for higher-quality vegetable seedlings. Government offices have promoted seedling use through free distribution, but gardeners appear willing to pay for better-quality seedlings. Commercial nurseries could potentially be encouraged to sell in urban areas. Community gardens, many of which are well-organized, could also help produce and sell seedlings to their communities. This could also present a business opportunity for community gardeners, and the government would likely support such an initiative.

Finally, in Dhaka, stakeholders also observed a substantial unmet market demand for high-quality vegetable seedlings. With appropriate external support, some rooftop gardeners could specialize in seedling production and sell seedlings within their communities. Such initiatives could create new economic opportunities for women or youth. Marketing seedlings through online platforms could also

be possible and create job opportunities for youth in delivery services.

3.4 Policy support for promoting urban and peri urban agriculture

The four cities share a commitment to supporting urban agriculture, with local governments implementing policies and programs to strengthen urban food production (Table 6). However, the level of institutionalization differs considerably. Metro Manila exhibits the most formalized support through the National Urban Agriculture Program, with Local Government Units routinely distributing free vegetable seedlings to community gardens, reducing reliance on commercial nurseries. Addis Ababa adopted a centralized governance model through the Urban Agriculture Commission, although seedling production remains largely informal and household-based. In Nairobi, support is embedded within county government programs and complemented by research and development partners; however, respondents consistently reported weak extension services despite the presence of multiple institutional actors. In Dhaka, support is driven mainly by the Department of Agricultural Extension and non-governmental organizations promoting rooftop gardening, with social media increasingly used for technical exchange and seedling marketing.

TABLE 6 Policy support for urban agriculture and seedling systems.

Policy domain	Nairobi (Kenya)	Addis Ababa (Ethiopia)	Dhaka (Bangladesh)	Metro Manila (Philippines)
Policy framework	Nairobi City County Food Systems Strategy (2022–2027); integrated urban resilience agenda	Urban Agriculture Commission-led framework	Rooftop gardening via national extension programs	National Urban Agriculture Program via Local Government Units
Lead institutions in urban agriculture	County department in agriculture; NGOs; research partners	Urban Agriculture Commission	Department of Agricultural Extension (DAE); NGOs; local partners	Local Government Units; Department of Agriculture
Financing & incentives	Project/donor-based; limited county funding; pilot innovation support	Moderate public support; limited private incentives	Donor/NGO dependent; weak public finance	Strong Local Government Unit funding; subsidized inputs; free seedlings
Coordination	NEFSALF and multi-stakeholder platforms (government, NGOs, research, development partners)	Urban Agriculture Commission coordination platform	NGO–DAE coordination networks; fragmented MSPs	Local Government Unit-led coordinated national implementation framework

Financing for seedling systems is directed primarily toward household and community production rather than commercial nursery development. Public programs mainly support demonstration gardens, training, and free seedling distribution, while private nursery operators rely largely on their own investment. Respondents in Nairobi and Addis Ababa identified limited access to affordable credit as a major constraint to nursery expansion. In contrast, Metro Manila's stronger public support reduces production costs for community gardeners but also weakens incentives for private seedling markets.

Institutional arrangements were characterized by multi-actor partnerships rather than single coordinating authorities (Table 6). In Nairobi, the Nairobi and Environs Food Security, Agriculture and Livestock Forum (NELSALF) provides a platform for coordinating government, research, NGO, development partner, and private-sector efforts. By comparison, Metro Manila demonstrates stronger vertical integration between national policy and Local Government Units. Despite these different arrangements, respondents across all four cities reported weak coordination between agriculture, urban planning, and environmental sectors, limiting policy attention to vegetable seedling systems.

4 Discussion

4.1 Urban vegetable production and seedling demand

Despite differences in geography, governance, and production systems, the four cities exhibit remarkably similar patterns of urban vegetable production. Production was dominated by small-scale, intensive systems cultivating primarily leafy vegetables, reflecting the need to maximize productivity on limited land while meeting growing urban demand for fresh produce. This pattern is consistent with studies showing that urban agriculture favors short-cycle, high-value crops suited to space-constrained

environments (Orsini et al., 2013; Hamilton et al., 2014). However, the predominance of a few exotic vegetables and limited integration of indigenous crops represent missed opportunities to improve dietary diversity, nutrition, and climate resilience. Although urban production contributes to local food availability, it remains insufficient to meet city demand because of persistent constraints, including limited land, pest and disease pressure, inadequate access to quality inputs, and weak extension services (Weidner et al., 2019; FAO et al., 2023b).

The predominance of leafy vegetables also reflects strong demand for quality seedlings. Crops such as kale, spinach, lettuce, and cabbage are well suited to small plots, container gardens, and rooftop systems because of their short production cycles, while fruiting vegetables such as tomato, eggplant, and pepper present opportunities for specialized and more commercial nursery enterprises, particularly where improved propagation methods, including grafting, can reduce soil-borne diseases and increase productivity.

4.2 Urban seedling systems: constraints and opportunities

Despite increasing demand, vegetable seedling systems remain a critical but underdeveloped component of urban food systems. Formal nurseries and certification systems serve only a small share of the market, with most urban producers relying on direct seeding or seedlings from informal nurseries that supply seedlings of variable quality. This reflects broader trends across sub-Saharan Africa and South and Southeast Asia, where informal input systems dominate because regulatory and institutional support has not kept pace with urban food system development (Orsini et al., 2013; Battersby et al., 2024). Seedling quality is constrained by open-field propagation, limited use of protected nursery technologies and quality growing media, weak technical capacity among nursery operators, and fragmented market linkages.

Common constraints included pest and disease pressure, particularly soil-borne pathogens affecting bare-root seedlings, limited technical training, and weak linkages between nurseries, retailers,

and growers. Eco-friendly technologies such as soilless media, grafted seedlings, and biodegradable containers also remained underutilized. Similar challenges have been reported elsewhere, where weak nursery infrastructure and poor market integration constrain adoption of improved planting materials (Schreinemachers et al., 2018). In Metro Manila, extensive public distribution of free seedlings may also reduce incentives for private-sector investment in commercial seedling production (Bayot et al., 2023).

Despite these constraints, substantial opportunities exist to strengthen seedling systems. Stakeholders across all four cities identified strong unmet demand for high-quality vegetable seedlings, particularly among household and small-scale producers. Quality seedlings are productive assets that improve crop establishment, shorten production cycles, increase resource use efficiency, and reduce pest and disease losses, enabling growers to maximize production on limited urban land. Tray-grown seedlings improve crop establishment under irregular watering and high temperatures, creating business opportunities for commercial and community-based nurseries. Expanding commercial nurseries, strengthening certification and quality assurance, promoting protected propagation technologies, and leveraging digital platforms for seedling marketing could improve access to quality planting materials. Greater commercialization of indigenous vegetable seedlings also offers opportunities to diversify diets, improve nutrition, and enhance climate resilience. Similar investments in seedling supply chains have strengthened food security and smallholder livelihoods (Coyne et al., 2024; Claereboudt et al., 2024).

4.3 Policy and institutional implications

Although all four cities recognize urban agriculture within broader food system agendas, policy support for vegetable seedling systems remains largely indirect. Existing programs emphasize household food production, nutrition, and community gardening, with comparatively limited attention to commercial seedling enterprises that underpin reliable access to quality planting materials. Institutional maturity alone does not necessarily translate into stronger seedling systems. Metro Manila benefits from a well-established policy framework and sustained public investment through Local Government Units, whereas Nairobi and Addis Ababa rely more heavily on partnerships among government agencies, research institutions, development organizations, and civil society. In Dhaka, digital platforms increasingly complement formal extension by facilitating technical advice and seedling marketing where public capacity is limited (Weidner et al., 2019; FAO et al., 2023b).

Across the four cities, financing remained a major constraint, with public investment focused largely on training, demonstration gardens, and free seedling distribution rather than commercial nursery development. Similar governance and financing gaps have been reported elsewhere, where urban agriculture policies prioritize production over input systems and agribusiness development (Hamilton et al., 2014; Weidner et al., 2019). Integrating commercial seedling enterprises into urban food system strategies, strengthening quality assurance and extension services, improving access to finance, and promoting public-private partnerships could enhance the availability of quality planting materials while supporting more productive, resilient, and nutrition-sensitive urban food systems (FAO et al., 2023b).

5 Conclusions and implications

High quality seedlings are the lifeblood of resilient and productive UPU farming systems. Yet, across Addis Ababa, Dhaka, Metro Manila, and Nairobi, seedling systems remain an overlooked weak link. This study underscores the urgent need for investment in seedling quality, market development, and institutional support to unlock the full potential of UPU agriculture. By adopting a strategic and inclusive approach, it is possible to build a future in which urban production systems flourish, cities gain reliable access to fresh vegetables, and farmers and growers thrive through robust, sustainable seedling systems.

Moving forward, the evidence from these four cities points to a multi-pronged approach for upgrading seedling systems in urban and peri-urban agriculture. Investment in capacity building is a foundational priority. Establishing accredited training programs on nursery management, seedling propagation, pest and disease management, and the use of soilless media would substantially raise the technical competency of informal nursery operators. Where possible, certification of propagators should be linked to accessing government support programs, creating an incentive structure for quality compliance.

Developing urban marketing pathways is equally important. Local governments are well placed to facilitate urban seedling markets, support market days, and create enabling environments for nursery operators to reach urban producers. Mobile applications and digital platforms could help link growers with suppliers, track seedling availability, and build consumer awareness of quality indicators, an approach especially promising in Nairobi and Dhaka, where mobile phone penetration is high.

Promoting technological innovation represents a further strand of action. Subsidizing the adoption of soilless propagation media, vertical farming technologies, and renewable or biodegradable packaging would lower entry barriers for urban propagators and align seedling production with climate-smart agriculture goals. Public-private partnerships are essential: linking government programs with private nurseries, agri-tech startups, and community gardens could create more diverse and robust distribution networks. Strengthening policy frameworks by revising urban agriculture and seed system policies to more explicitly integrate seedling production and marketing would help ensure alignment with broader sustainable development goals. Finally, ongoing research and data collection, including longitudinal monitoring of seedling performance, market dynamics, and consumer preferences, will be vital for tracking system improvement and adapting interventions as urban contexts evolve. If these interventions are pursued, seedling systems in the four focus cities could evolve from informal, fragmented structures into organized, sustainable contributors to urban food security and resilience by 2030.

6 Limitations

Several limitations of this study should be acknowledged. The analysis relied on a descriptive thematic synthesis of stakeholder interviews and city-level scoping reports rather than a formal comparative analysis with statistically representative sampling. Stakeholders were identified using purposive and snowball sampling to ensure value chain representation, which introduces a

potential selection bias and limits generalizability to the broader population of urban food producers. The scoping studies varied slightly in scope and depth across cities, reflecting differences in the available evidence base and local contexts, making direct cross-city comparisons difficult. Reliance on self-reported data from stakeholder interviews also introduces the possibility of social desirability bias, particularly in responses related to regulatory compliance and seedling quality. The study was conducted between 2022 and 2024, and some city-specific conditions, including policy developments and market structures, may have changed. Future research with longitudinal designs, larger representative samples, and standardized data collection instruments would strengthen the evidence base for seedling system interventions in urban contexts.

Data availability statement

The raw data supporting the conclusions of this article will be made available by the authors, without undue reservation.

Ethics statement

Ethical approval was not required for the study involving humans because it consisted solely of general interviews using non-personal, non-sensitive questions and therefore did not require prior ethics review. The studies were conducted in accordance with the local legislation and institutional requirements. The participants provided their written informed consent to participate in this study.

Author contributions

DC: Supervision, Conceptualization, Writing – review & editing, Data curation, Formal analysis, Project administration, Funding acquisition. EN: Writing – original draft, Writing – review & editing. JA: Writing – original draft, Writing – review & editing. FO: Writing – review & editing, Methodology, Investigation. MA: Formal analysis, Writing – review & editing, Methodology, Investigation. SK: Formal analysis, Writing – review & editing, Methodology, Investigation. EA: Writing – review & editing, Formal analysis, Methodology, Investigation. ZA: Investigation, Formal analysis, Writing – review & editing, Methodology. WB: Formal analysis, Investigation, Methodology, Writing – review & editing. PS: Writing – original draft,

Supervision, Funding acquisition, Validation, Writing – review & editing.

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Conflict of interest

The author(s) declared that this work was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

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The author(s) declared that Generative AI was not used in the creation of this manuscript.

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